Priority of operations with positive and negative integers and decimals

Task A: Priority of operations with negatives

1) Fill in the blanks to make the calculation in the centre 24

24

$60 + (-12) \times 3$

$(-240) \div (-10)$

$4 + (-2) \times (-10)$

$20 - (-8) \div 2$

$40 + (8) \times (-2)$

2) Work out the following, ensuring to show your working out.

a)
$$100 + \sqrt{100 - (-21)} \div (-11)$$
$$100 + \sqrt{100 + 21} \div (-11)$$
$$100 + \sqrt{121} \div (-11)$$
$$100 + 11 \div (-11)$$
$$100 + \frac{11}{(-11)}$$
$$100 + (-1) = 99$$

b)
$$50 - (-100 + 149) \div (-7)$$
$$50 - (149 + (-100)) \div (-7)$$
$$50 - (49) \div (-7)$$
$$50 - \frac{49}{(-7)}$$
$$50 - (-7)$$
$$50 + 7 = 57$$

3) Fill in the pyramid. The number above the block is the sum of the two below.

Top: $(-23) + 16 = -7$

Middle: $(-37) + 14 = -23$ | $14 + 2 = 16$

Bottom:

$\frac{90}{(-2)} + \frac{4^2}{2}$	$20 + \frac{(-4)}{(-2)} \times (-3)$	$4 - \frac{\sqrt{100}}{(-10)} \times (-2)$

Task B: Priority of operations with negatives and decimals

1) Work out the following, ensuring to show all your working out.

$$3.4 - 7.6 + 11.2$$
$$3.4 + (-7.6) + 11.2$$
$$14.6 + (-7.6) = 7$$

2) Work out the following, ensuring to show all your working out.

$$13.45 - 2.4 + 3.2 - 1.6$$
$$13.45 + (-2.4) + 3.2 + (-1.6)$$
$$16.65 + (-4) = 12.65$$

3) Work out the following, ensuring to show all your working out.

$$184.2 - (-45.2) \times 16 \div (-4)$$
$$184.2 - (-45.2) \times \frac{16}{(-4)}$$
$$184.2 - (-45.2) \times (-4)$$
$$184.2 - 180.8 = 3.4$$

4) Work out the following, ensuring to show all your working out.

$$10.2 - 1.4 \times (-3.2) + \frac{12^2}{\sqrt{25 + 75}}$$
$$10.2 - 1.4 \times (-3.2) + \frac{12^2}{\sqrt{25+75}}$$
$$10.2 - 1.4 \times (-3.2) + \frac{12^2}{\sqrt{100}}$$
$$10.2 - 1.4 \times (-3.2) + \frac{12^2}{10}$$
$$10.2 - 1.4 \times (-3.2) + \frac{144}{10}$$
$$10.2 + 4.48 + 14.4 = 29.08$$

5) Create your own two questions using a combination of operations and decimals so the correct answer is 24. You must use decimals and ensure to check your calculation with a calculator. *There are an infinite number of calculations. These will need to be checked.*